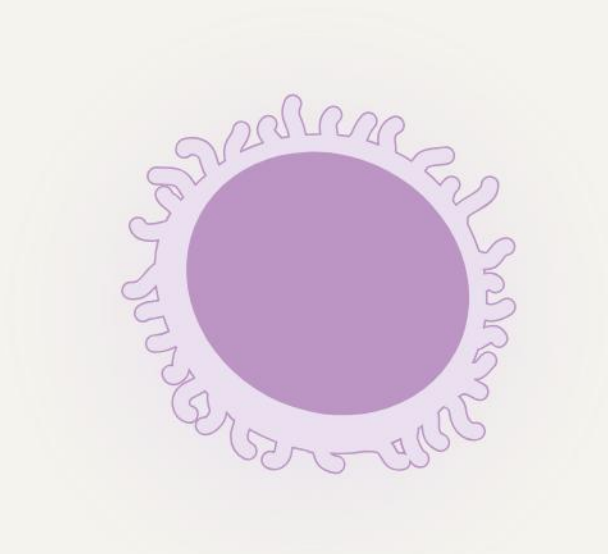


CASE STUDY 03

Calaspargase pegol

A smaller, smarter trial in adult Ph-negative ALL

A pharmacometrics-anchored design — and an FDA-agreed 36% enrollment reduction in a rare adult oncology population.



LYMPHOCYTE · CELL OF ORIGIN

The lymphoblast clone in adult Ph-negative ALL

COMPOUND	POPULATION	AGENCY	OUTCOME	VERDICT
Calaspargase pegol (Asparlas)	Adult Ph-negative ALL	FDA · Type A · 21 Jul 2023	N = 60 agreed (94 → 60 · -36%)	N = 60 AGREED

FDA Type A meeting · 21 Jul 2023 · NCT04817761 (SPARK-ALL) · Lymphocyte diagram © Cancer Research UK / Wikimedia · CC BY-SA 4.0

03 / 03

Approved in pediatrics. *Adults needed a smarter design.*

Same drug, same biology, same FDA-validated NSAA surrogate. The constraint wasn't scientific doubt — it was *operational feasibility*. Ninety-four was deliverable in the protocol and undeliverable in practice.

THREE NUMBERS THAT FRAME THE PROBLEM

2018

FDA PEDIATRIC APPROVAL

Ages 1 mo – 21 yr · NSAA surrogate · 2,500 U/m² q21d

94

ORIGINAL SAMPLE SIZE

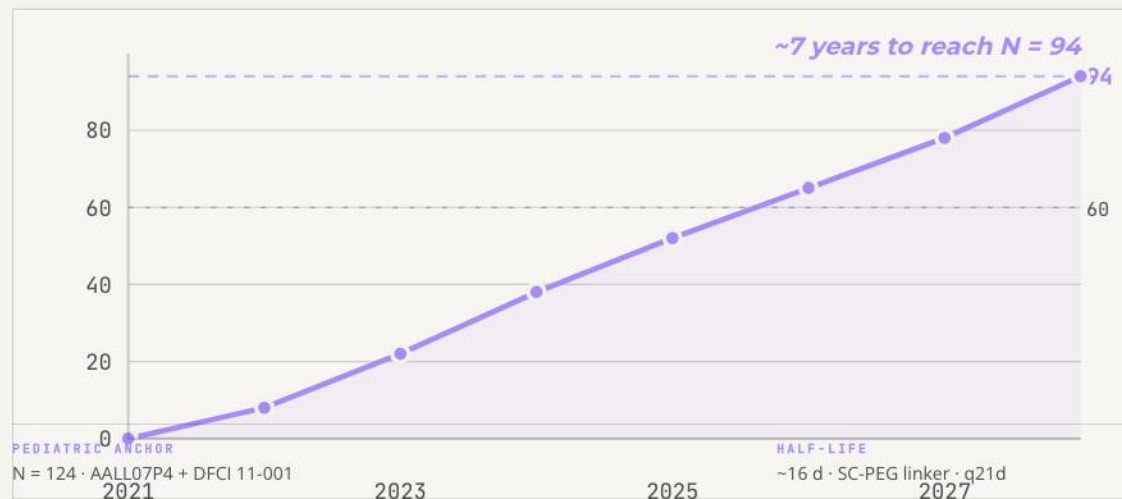
Endpoint-powered · target lower 95% CI ≥ 90% NSAA achievement

~2028

IF DESIGN UNCHANGED

SPARK-ALL projected enrollment under endpoint-powered design

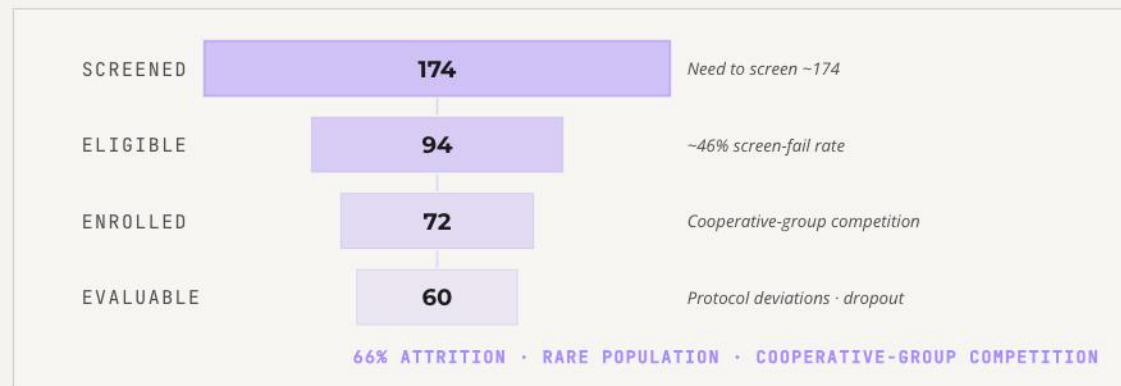
PROJECTED CUMULATIVE ENROLLMENT



CASE 03 · CHALLENGE

Source · FDA label 761102 (Dec 2018) · NCT04817761

PATIENT ATTRITION — WHY 94 TAKES UNTIL ~2028



Could a **smaller, smarter study** still be defensible to FDA?

The drug worked in pediatrics. The pediatric PopPK model was **FDA-reviewed and label-supporting**. The question was whether the same scientific question could be answered with **fewer adults and more model**.

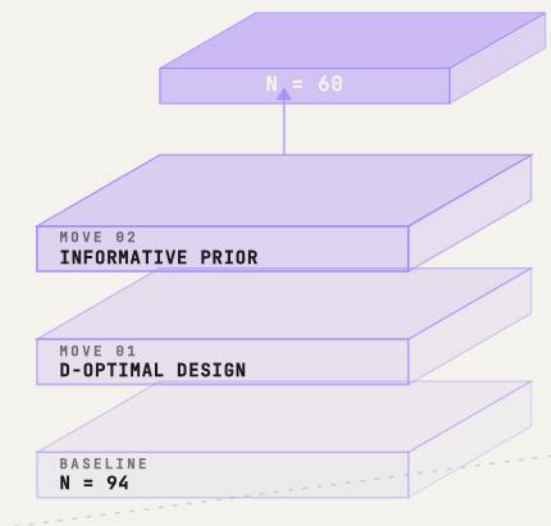
TRIAL

SPARK-ALL · NCT04817761

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Two innovations, *individually precedented* — stacked for adult oncology.

Each move is *FDA-precedented on its own*. Combined here for the first time in this clinical context.



THE RESULT

N = 60 —36% vs baseline · FDA-agreed at Type A · 21 Jul 2023.

02 MOVE #2 · INFORMATIVE PRIOR

Adult data anchors — does not re-derive.

Pediatric PopPK · N = 124 (AALL07P4 + DFCl 11-001)

Adult sample augments the model · model is not re-fit.

FDA PRECEDENT · RYLAZE (ADULT, 2021) · ASPARLAS (PEDIATRIC, 2018)

01 MOVE #1 · OPTIMAL DESIGN

Anchor to PK precision — not endpoint power.

Fisher Information Matrix · D-optimality

Detect ±20% CL difference adult vs pediatric.

FDA PRECEDENT · MENTRÉ · BORNKAMP · PINHEIRO

BASELINE

Conventional adult oncology · **N = 94**

What an endpoint-powered trial in this rare adult population would otherwise require.

THE STACK PK-precision sample size + informative pediatric prior — two FDA precedents combined for the first time in this clinical context, returning the same evidence quality at a smaller sample.

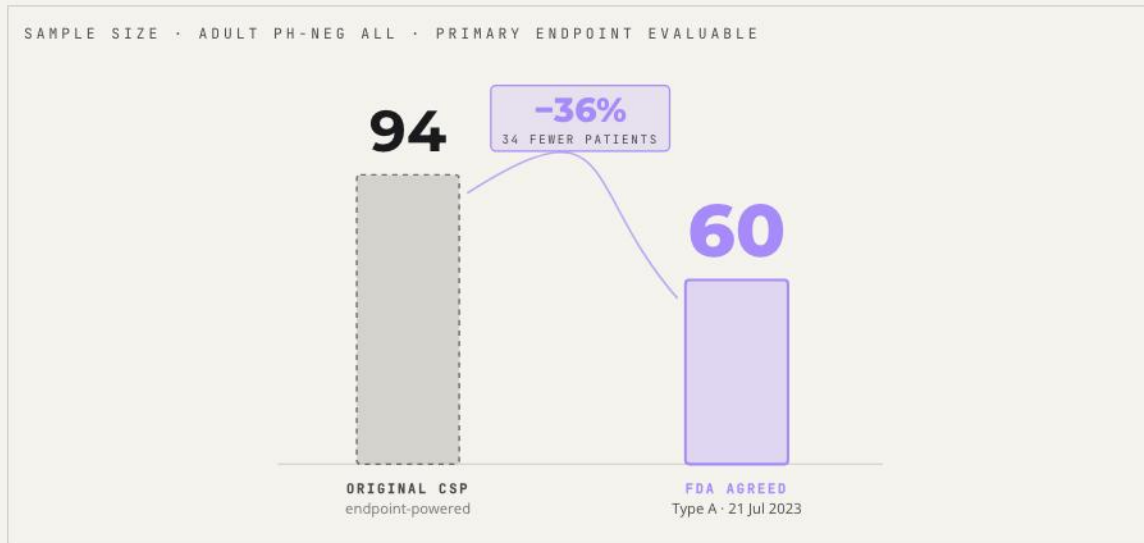
CASE 03 · STRATEGY

Source · PopED workflow (Tessier / Riglet, Paris) · Mentré et al. · NCT04817761

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FDA agreed to **N = 60** — a 36% reduction in adult enrollment.

FDA Type A · 21 July 2023 · pharmacometrics-anchored briefing in a rare adult oncology population — *three of four pillars agreed on the record.*



PILLARS AGREED · PILLARS REPOSITIONED

3 / 4 PILLARS AGREED
on the FDA Type A record

> 85% AE-DETECTION PROBABILITY
safety framework anchored — prepared in parallel with optimal design

Repositioned SIMULATED PRIMARY
to dose confirmation in Cohorts 1 & 2 · Part 2 pending additional PopPK

REGULATORY POSITION · TYPE A MEETING RECORD

Sixty patients delivered the >85% AE-detection probability the agency was looking for — the same threshold the safety framework was sized against.

— FDA TYPE A · 21 JUL 2023

The simulated primary was not rejected — repositioned. FDA required additional PopPK in Cohorts 1 & 2 before Part 2 could rely on it.

— FDA TYPE A · 21 JUL 2023

PHARMACOMETRIC METHODOLOGY Two frameworks prepared · one sample size agreed — *the methodology now travels independent of trial outcome.*

CASE 03 · FDA ENGAGEMENT

Source · FDA Type A meeting · 21 Jul 2023 · NCT04817761

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Sixty adults anchor a model that *already knows* most of the answer.

At $N = 60$ the model is *as precise as at $N = 94$* for the parameters that drive dose decisions — because the information lives in the pediatric prior, not the adult sample alone.

WHY $N = 60$ IS ENOUGH · TWO EVIDENCE BLOCKS

01 THE PEDIATRIC ANCHOR

The adult data test the model — they don't rebuild it.

- ▶ $N = 124$ pooled pediatric PopPK · saturable distribution · combined linear/non-linear elimination
- ▶ ~16-day half-life · BSA / age / sex covariate structure
- ▶ FDA-reviewed · label-supporting (Asparlas pediatric, 2018)
- ▶ Adult Part 1 external VPC vs pediatric-model predictions: no structural failure



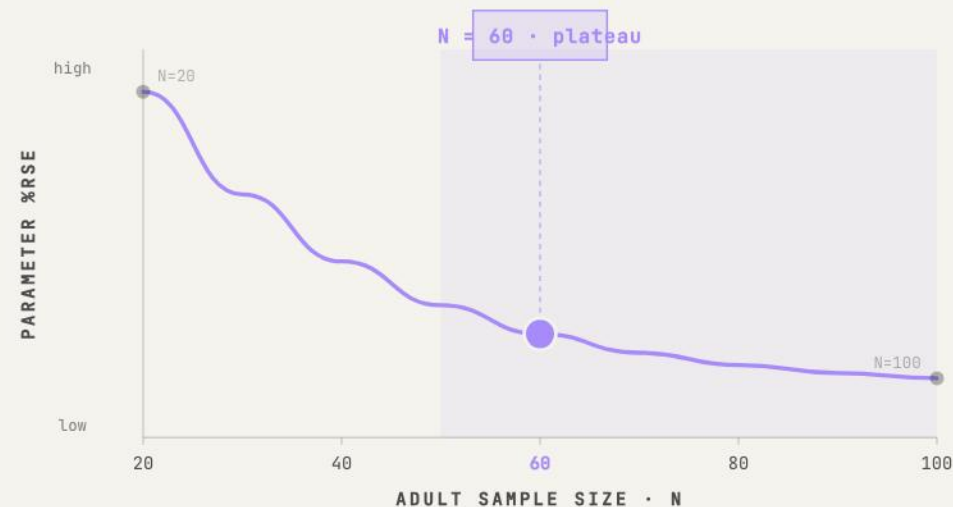
02 SENSITIVITY ANALYSIS (ILLUSTRATIVE)

Three independent tests — all say sixty is enough.

- ▶ Cohort-1:Cohort-2 enrollment ratio varied across scenarios → conclusion stable
- ▶ %RSE of key PopPK parameters not sensitive to adult N above $\approx 50-60$
- ▶ Bootstrap replicates · pcVPC on pooled data → predictive performance unchanged



PARAMETER %RSE VS ADULT SAMPLE SIZE · PLATEAU BEYOND $N = 50-60$



illustrative · not a specific parameter

The smaller sample size doesn't weaken the science — it *clarifies where the scientific evidence actually lives*.

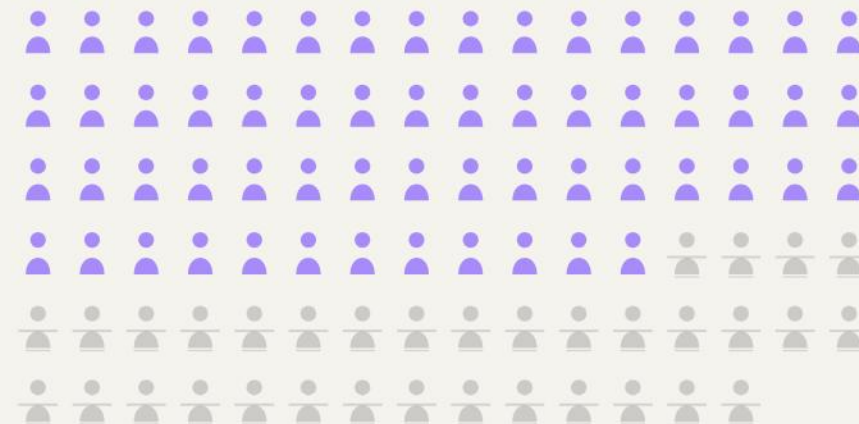
A **36% enrollment reduction**. A documented precedent. A methodology that travels.

FDA-agreed at Type A · pharmacometric anchor + safety framework — the same N landed by two independent methods, with **34 patients not enrolled**.

-36%

SAMPLE-SIZE REDUCTION · ADULT PH-NEG ALL

From **94** patients budgeted to **60** required
— primary-endpoint evaluable, FDA-agreed
via AE-detection probability $\geq 85\%$.



■ 60 ENROLLED ■ 34 NOT ENROLLED · EACH FIGURE = ONE PATIENT

REGULATORY

Type A

FDA Type A record of a 36% reduction in adult enrollment · anchored on AE-detection probability $\geq 85\%$.

OPERATIONAL

34 patients

Per-patient information density up · trial footprint down · faster path to read-out.

SCIENTIFIC

2 → 1

Two independent statistical paths (pharmacometrics + biostatistics) converged on the same N.

DURABLE



Methodology now travels independent of the trial outcome — template for rare-population designs.

DURABLE CONTRIBUTION SPARK-ALL ended at N = 42 on a sponsor portfolio decision (Feb 2026), independent of design quality — the FDA-agreed methodology is durable beyond any single program.

The methodology *scales* — and the value isn't one trial. It's a template.

THE TEMPLATE

Generalizes wherever a reference population is well-characterized and observed-endpoint trials aren't feasible.

- 1 Anchor sample size on parameter precision · not endpoint power
- 2 Use a model-based simulated primary when observation isn't feasible
- 3 Stack two FDA-precedented methods · don't bet on one untested move
- 4 Quantify the safety framework alongside the design — agencies anchor on it
- 5 Brief the regulator with the methodology, not just the number

WHERE IT GOES NEXT — MERCK CMD PIPELINE

Enlicitide decanoate · oral PCSK9 · CORALreef LDL-C reduction · the same **D-optimal design under informative prior** applies wherever Phase 3 dose justification rides on a small precision-critical sample and a well-characterized class precedent.

Beyond Enlicitide: *HRS-5346 · MK-2060 · next-generation class extensions* — the template is direct application, not adaptation.

CLOSING · BREADTH + RECORD AT SCALE

The win wasn't a smaller trial — it was a precedent the next program can cite.

ICH M15 · FEB 2026 First global guideline on **model-informed drug development** — FDA documented this case in 2023 · M15 codifies the playbook in 2026.

FRAMEWORK THEMES IN THIS CASE STUDY · 3 OF 5

01 ⇄ QP REPLACES STUDY

A simulated primary substituted for an endpoint-powered $N = 94$ adult trial.

04 ◇ NOVEL METHODS

Two FDA-precedented methods stacked — first time in adult oncology.

05 ◇ JUDGMENT

Type A briefed pre-execution · CI relaxed 90% → 85% · pillar repositioned to keep the reduction.